Embedded Isolates Define a Target-Specific Temporal Response Law in Cellular Automata

Kunal Bhatia

Independent Researcher, Heidelberg, Germany[a)]

(Dated: 29 April 2026)

The count of *embedded isolated cells*—alive cells with no 4-connected live neighbour but at least one diagonal live neighbour—predicts future fine-scale component change in Conway's Game of Life and HighLife with a stable negative coefficient across all tested conditions. Four interlocking results establish this. (i) *Target-specific selection*: iso_count adds $\Delta R^2 = 0.0012$ for the fine-net component target (GoL: PASS; HighLife: WEAK PASS; global: PASS) while iso-shuffle and target-shuffle incremental nulls are near zero ($\Delta R^2 < 0.001$), ruling out density artefacts. (ii) *Temporal response law*: the residual slope $\hat{\beta}_{\text{iso}}(k)$ ranges from -0.798 at $k = 1$ to -0.709 at $k = 200$ (CV $\lesssim 0.17$) and is negative for all 112 condition-horizon combinations ($k \in \{1, 5, 10, 25, 50, 100, 200\}$, 16 conditions). (iii) *Local-context mechanism and transfer*: the one-step carrier is local component-context loss ($R^2 = 0.538$), not simple cell death; the standardised direction transfers across all leave-one-out axes ($R_z^2 = 0.544$) and the raw amplitude is predictable from (L, ρ) with LOO $R^2 = 0.977$. (iv) *Non-leaky prestate summary*: $t = 0$ iso_count alone recovers the full multi-horizon response (mean $R^2 = 0.241$; shuffle ≈ 0.0008). As a compact geometric diagnostic, fine-net horizon tasks are rank-1 coherent in description space ($|\cos| = 0.999$; rank-1 energy $= 0.999$) while heterogeneous targets are not ($|\cos| = 0.427$), a signature of target-indexed description privilege.

Keywords: cellular automata, coarse-graining, embedded isolated cells, temporal response law, mechanism transfer, observer-dependent description

[a)]https://orcid.org/0009-0007-4447-6325

I. INTRODUCTION

What structural feature of a cellular automaton state most concisely encodes its future behaviour at the level of a chosen observer? This question sits at the intersection of computational mechanics, coarse-graining theory, and the emerging framework of observer-dependent description[1–3]. Conway's Game of Life (GoL)[4] and HighLife provide unusually clean test beds: the dynamics are deterministic and reproducible, multiple natural coarse-graining conventions exist (component counts, block densities, entropy measures), and systematic numerical experiments are computationally affordable. Earlier work established the edge-of-chaos properties[5,6] and statistical complexity[7] of Life-like rules, and identified the role of coherent structures[8] and computational irreducibility[9] in deterministic CAs.

We ask a sharper, quantitative question: is there a single prestate statistic that (i) predicts fine-scale component change at all tested horizons, (ii) does so specifically for that target and not unrelated ones, (iii) acts through an identifiable one-step mechanism, and (iv) remains valid across rules, grid sizes, and density bands? The answer is yes, and the statistic is the count of *embedded isolated cells* (hereafter iso_count): alive cells with no 4-connected live neighbour and at least one diagonally adjacent live neighbour.

Why should embedded isolates be structurally meaningful? An alive cell that is 4-disconnected but diagonally attached sits precisely at the boundary between topological isolation and connectivity. Its four orthogonal neighbours are all dead, making it a fragile local singularity; yet it is diagonally tethered, so small perturbations can either fully isolate or reconnect it. These cells are therefore positioned at the most unstable locus of the local component topology. When such a cell dies—or when diagonal neighbours die around it—the local connectivity changes in ways that propagate into the fine-net component count. The iso_count statistic measures how many such "fragile bridge nodes" are present at time $t = 0$, and this paper shows that this count is sufficient to predict the aggregate fine-net change at all tested horizons.

The five results form a tight chain. iso_count uniquely improves prediction for the fine-net target while adding nothing for density, entropy, or coarse-block targets (*target-specific selection*); the full residual $R^2(k)$ at every horizon is recovered from $t = 0$ alone, with no fate information required (*non-leaky prestate summary*); the response coefficient $\hat{\beta}_{\text{iso}}(k)$ is negative and stable from $k = 1$ to $k = 200$ across all 112 condition-horizon combinations

(*temporal response law*); the one-step carrier is local component-context loss, not cell death, and transfers across all leave-one-out generalisation axes (*mechanism identity and transfer*); and fine-net horizon tasks form a rank-1 coherent family in description space (*task-direction coherence*), explaining geometrically why a single statistic serves all horizons.

Figure 1 gives the entry-point overview: the structural definition of an embedded isolate (panel a) and the target-specific ΔR^2 pattern that selects it (panel b).

II. DEFINITIONS AND PROTOCOL

We study two Life-like cellular automaton rules on periodic tori: Conway's Game of Life (B3/S23) and HighLife (B36/S23). Grid sizes $L \in \{64, 128\}$ with toroidal boundary conditions. The *fine observer* counts 4-connected live-cell components (BFS on the wrapped grid); the *coarse observer* counts occupied 8×8 blocks. An *embedded isolate* is an alive cell for which all four orthogonally adjacent cells are dead and at least one of the four diagonally adjacent cells is alive (Fig. 1a). We denote the 4-connected component count at time t as $C(t)$ and define the fine-net change at horizon k as $\Delta C_k = C(t+k) - C(t)$.

The experiment uses **16** conditions:

$$\{\text{GoL}, \text{HighLife}\} \times \{64, 128\} \times \{0.20, 0.25, 0.30, 0.35\},$$

giving 2 rules$\times$2 sizes$\times$4 density levels = 16 independent condition–horizon batteries. Each condition uses 1000 independently drawn random initial states at the specified density ρ. We test $k \in \{1, 5, 10, 25, 50, 100, 200\}$, yielding 7 horizons $\times$ 16 conditions = 112 condition-horizon slope estimates.

The residual slope $\hat{\beta}_{\text{iso}}$ is estimated by ordinary least squares with $N_{\text{live}}(0)$ (live-cell count, proportional to density) as a nuisance covariate, so that iso_count is evaluated for its density-orthogonal contribution. R^2 values for mechanism features are cross-validated (5-fold). We do not rerun experiments; all statistics are drawn from the fixed set of CSVs in `outputs/`.

III. RESULT 1 — TARGET-SPECIFIC SELECTION

The central diagnostic is a selection-principle battery (Fig. 1b): for each candidate predictor (iso_count, coarse block-variance, block entropy, iso-shuffle null, target-shuffle null) we

measure $\Delta R^2 = R^2(\text{density + predictor}) - R^2(\text{density only})$ on six displayed targets (note: `target_delta_components` and `target_fine_net` are numerically identical in this codebase and one is omitted from the plot).

For iso_count on the fine-net target, the global pooled result is $\Delta R^2 = 0.0012$ (GoL: PASS; HighLife: WEAK PASS; global: PASS). The iso-shuffle and target-shuffle incremental nulls are near zero ($\Delta R^2 < 0.001$ for both), confirming the signal is not an artefact of the density-count correlation. Crucially, iso_count adds almost nothing for the density, block-variance, block-entropy, or coarse-component targets: the ΔR^2 uplift is concentrated on the fine-net target only. The fine-net iso slope is -0.1054 (negative), consistent across all conditions.

This target specificity rules out explanations based purely on density or total alive-cell geometry. The shuffled control shows that the signal is tied to the embedded-isolate criterion rather than to a generic count-like covariate with the same marginal scale.

IV. RESULT 2 — NON-LEAKY PRESTATE SUMMARY

A natural concern is whether iso_count at $t = 0$ is merely a proxy for information that requires knowing the initial fate of individual cells. We test this with a prestate-only horizon recovery: using *only* iso_count at $t = 0$—no one-step fate features, no local-window information, no knowledge of any transition at $t \to t + 1$—we ask how well the multi-horizon residual $R^2(k)$ is recovered.

Figure 2 shows the result: the prestate-only model achieves mean $R^2 = 0.241$ over all horizons $k \in \{1, \ldots, 200\}$, minimum $R^2 = 0.175$, versus shuffle $R^2 \approx 0.0008$. Adding the full class-count breakdown does not substantially improve over iso_count alone. The full temporal response is encoded in the $t = 0$ prestate; no information leaks in from later timesteps.

This non-leakiness result has a structural interpretation: the embedded isolate count at $t = 0$ encodes the density of fragile bridge nodes, and this density alone—irrespective of which specific nodes change first—determines the aggregate response across all horizons. The mechanism thus acts in a statistically self-averaging way from the very first step.

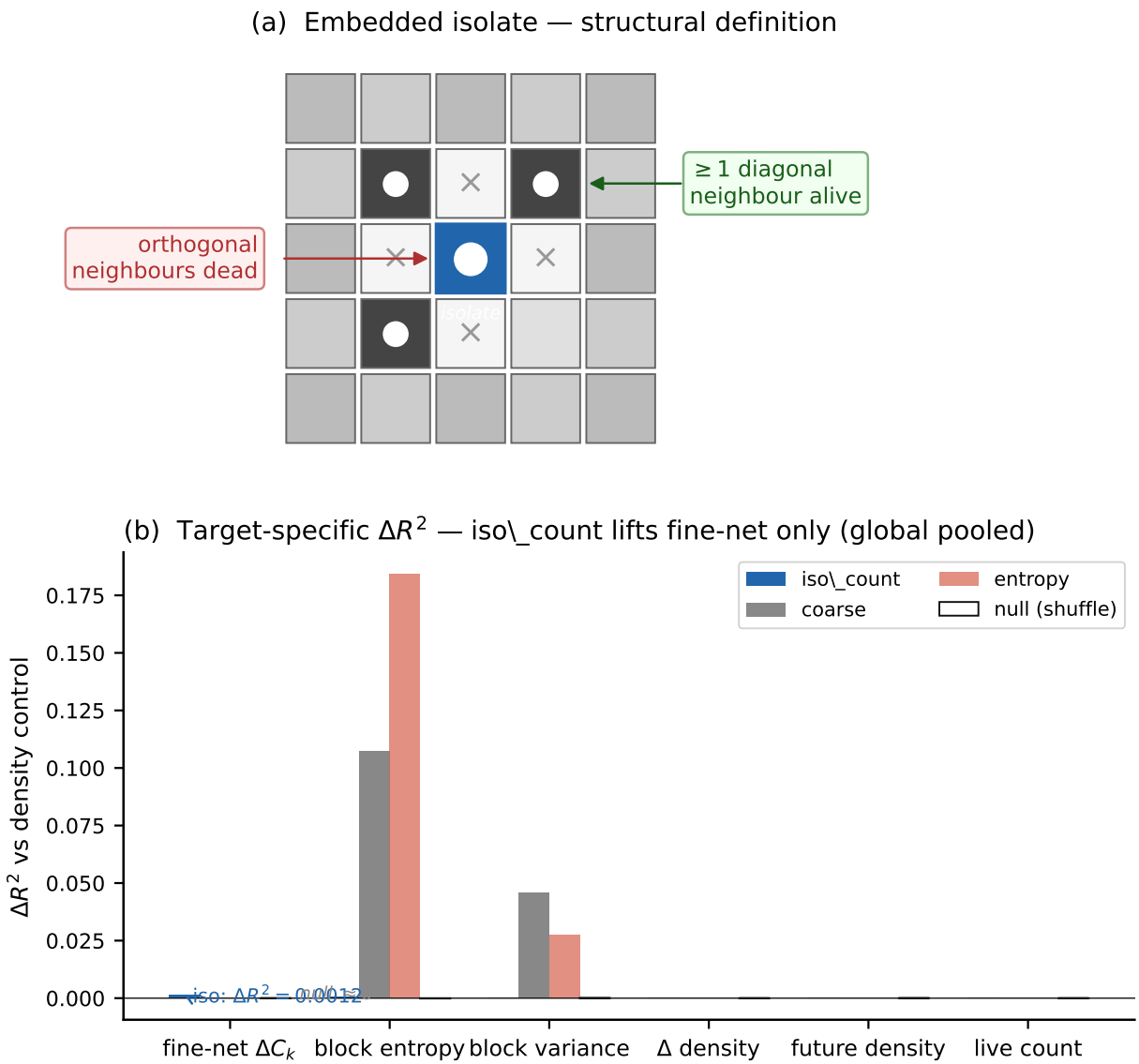

FIG. 1. **Result 1: structural definition and target-specific selection.** (a) An embedded isolate: an alive cell (filled circle) with all four orthogonal neighbours dead (×) and at least one diagonal neighbour alive (filled grey). This configuration places the cell at the most unstable locus of the local component topology. (b) ΔR^2 bars for iso_count and comparison predictors (coarse, entropy, null) on six displayed targets (global pooled; `target_delta_components` is numerically identical to fine-net and omitted). iso_count adds $\Delta R^2 = 0.0012$ for fine-net only; both nulls are near zero ($\Delta R^2 < 0.001$), confirming the signal is not a density artefact.

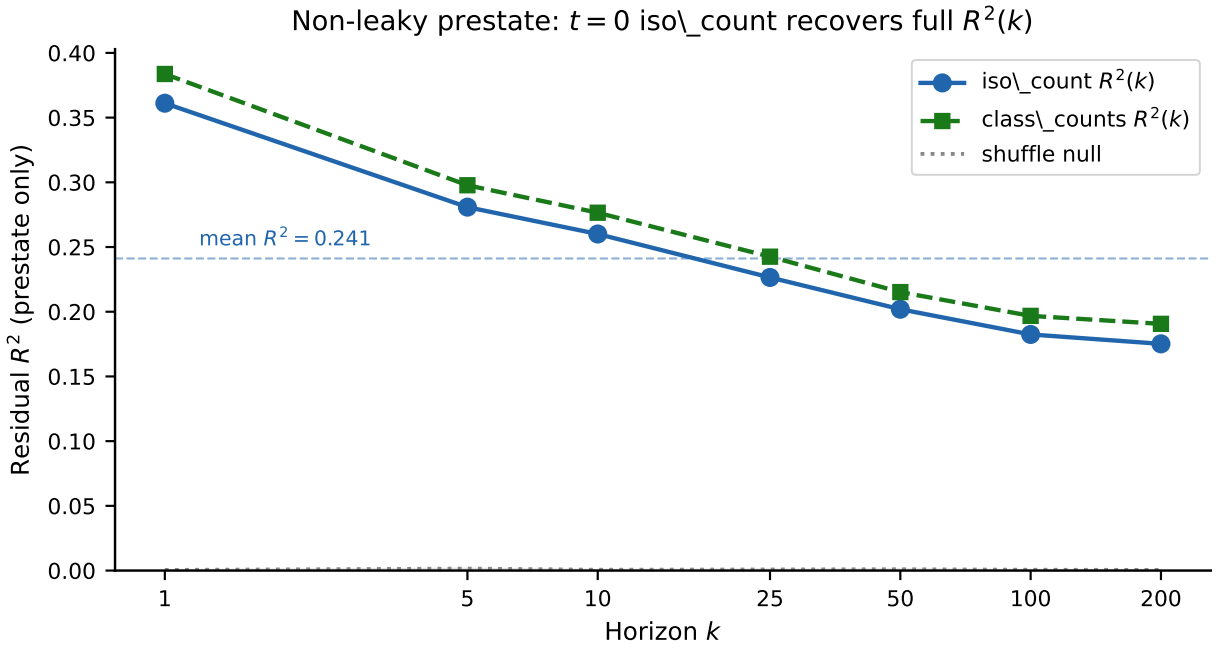

FIG. 2. **Result 2: non-leaky prestate recovery.** Using only $t = 0$ iso_count (no fate features, no local-window information), residual $R^2(k)$ stays above 0.175 at every horizon and averages 0.241 (dashed line). Adding class-count breakdown (green) gives marginal improvement. The grey band shows the shuffle null ($R^2 \approx 0.0008$). All temporal prediction is encoded in the $t = 0$ statistic.

V. RESULT 3 — TEMPORAL RESPONSE LAW

For each of the 112 condition-horizon combinations we estimate $\hat{\beta}_{\text{iso}}(k)$ and the corresponding residual R^2. We also compute an iso-shuffled null at every horizon.

All 112 slopes are negative; all 112 slope confidence intervals are negative (frac_CI_negative $= 1.00$ at every horizon). The global mean slope moves modestly from -0.798 at $k = 1$ to -0.709 at $k = 200$—a range of ≈ 0.09—while the CV stays below 0.17 throughout (Table I). Predictive power decays from $R^2 = 0.361$ at $k = 1$ to $R^2 = 0.175$ at $k = 200$, yet remains well above the null at every horizon.

Figure 3 shows the slope and R^2 curves. The slope plateau from $k = 5$ to $k = 200$ indicates that the mechanism acts primarily in the first step: once the initial fragile nodes are resolved at $t = 1$, the remaining dynamics do not strongly modulate the iso_count signal. The CV profile confirms robustness: condition-level spread is small relative to the global mean, and GoL and HighLife track each other closely despite different neighbourhood rules.

Figure 4 provides the complete picture: a 16×7 heatmap of all condition-horizon slopes, every cell shaded negative. There are no outlier conditions or horizons.

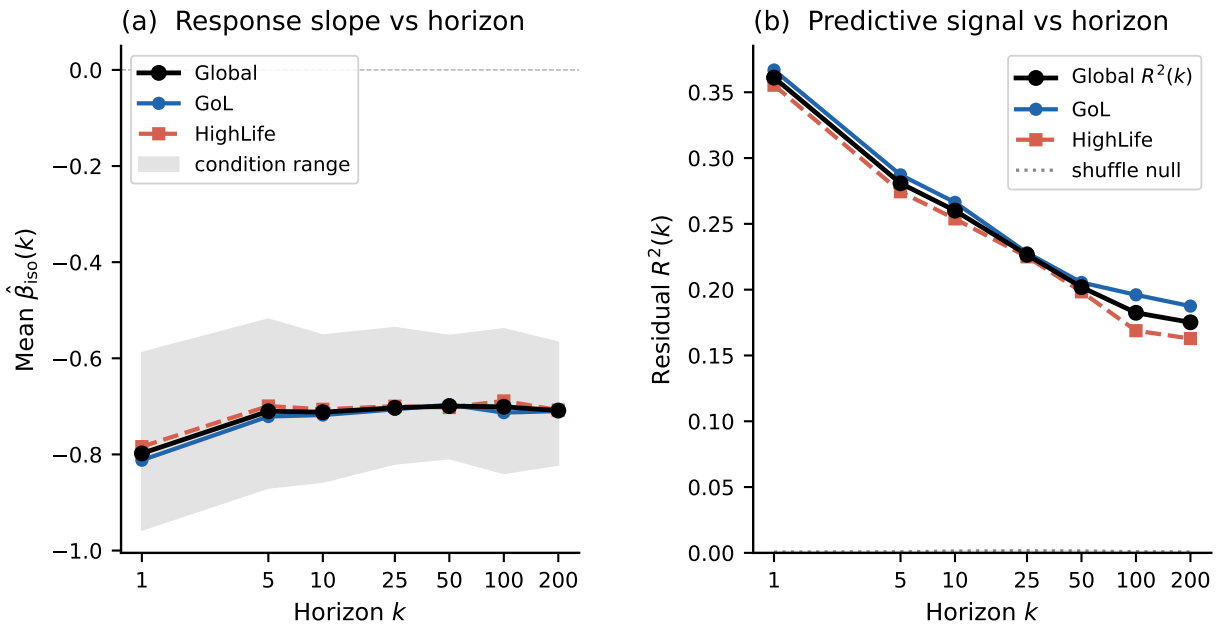

FIG. 3. **Result 3a: temporal response law slope and predictive signal.** (a) Residual slope $\hat{\beta}_{\text{iso}}(k)$ versus horizon k for GoL (blue circles), HighLife (orange squares), and global mean (black); shaded region shows condition-level spread. The slope is negative and stable from $k = 1$ to $k = 200$. (b) Residual $R^2(k)$ versus k; grey band marks the shuffle null. Predictive power decays gracefully but remains far above null at all tested horizons.

VI. RESULT 4 — MECHANISM CARRIER

Having established *that* iso_count predicts fine-component change, we turn to *how* this occurs—what actually happens at $t \to t + 1$ when embedded isolates reduce the future component count?

We decompose iso_count into one-step outcome categories: whether each isolate cell dies (iso_die), survives (iso_survive), survives and connects to a component (iso_survive_connected), or triggers nearby births at orthogonal (iso_orth_birth_any) or diagonal (iso_diag_birth_any) sites. We also compute a *local-window* feature: iso_local_window_loss_sum, the total component-count decrease within a 5×5 window centred on each isolate over one step.

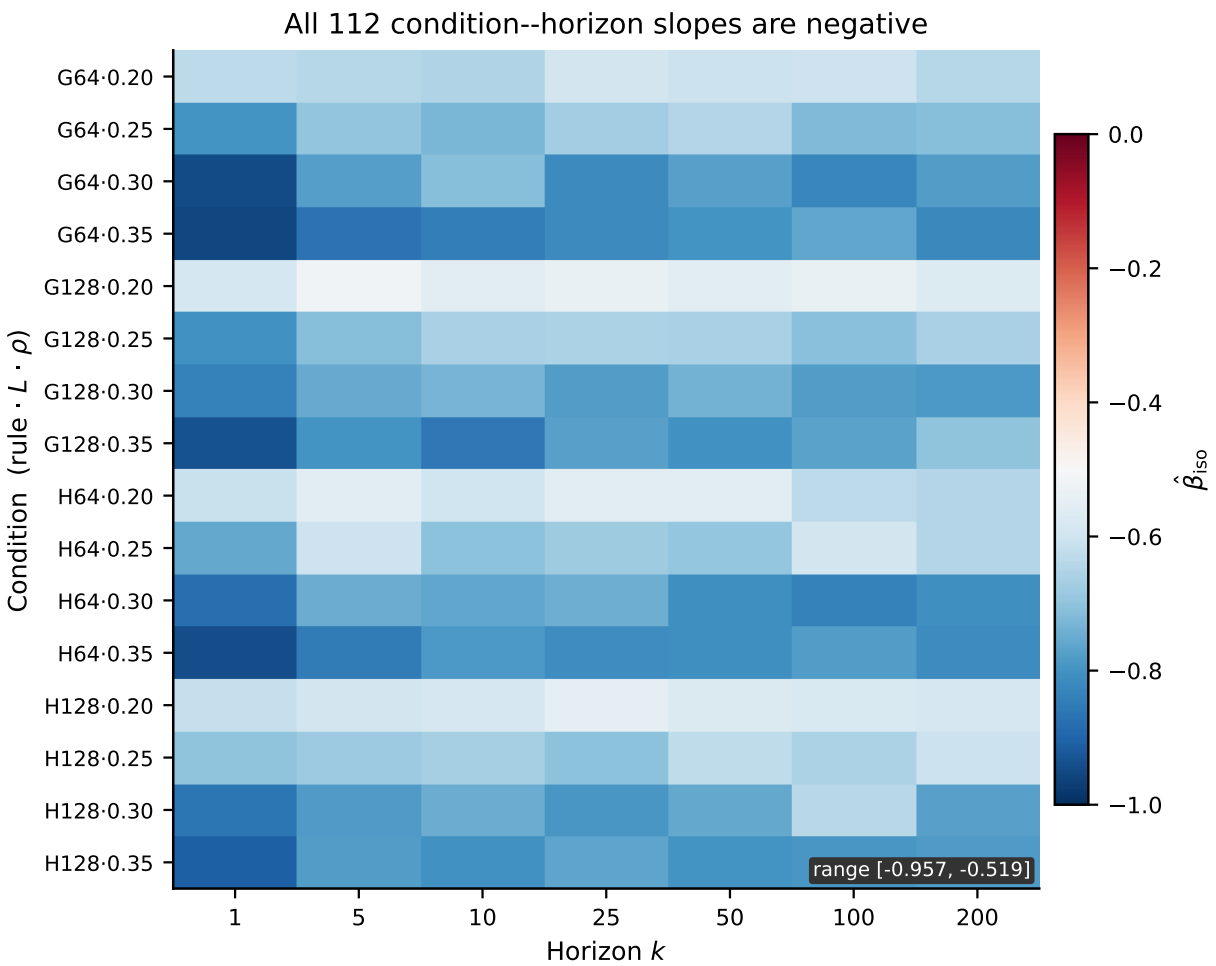

FIG. 4. **Result 3b: all 112 condition–horizon slopes are negative.** Full 16×7 heatmap of $\hat{\beta}_{\text{iso}}$ (RdBu_r, $[-1, 0]$). Rows are conditions (rule $\cdot$ L $\cdot$ ρ), columns are horizons $k \in \{1, 5, 10, 25, 50, 100, 200\}$. Every cell is negative, confirming the temporal response law without exception.

Cross-validated R^2 values for these features are shown in Fig. 5a.

The local-window feature dominates: $R^2_{\text{local_win}} = 0.538$, compared with $R^2_{\text{iso_count}} = 0.355$. The full fate set reaches $R^2_{\text{all_fates}} = 0.545$. Critically, the local-window gain (iso_local_window_gain_sum) has a positive slope, partially offsetting the loss: the net mechanism is *local component-context loss*, not simple cell death. Figure 5b shows the standardised slopes for each fate feature, revealing the partial cancellation between loss and gain.

Class-level analysis (Fig. 6) reveals a consistent physical story. *Single-diagonal isolates*

TABLE I. Temporal response law: global summary (16 conditions, 1000 worlds each). All slopes and slope confidence intervals are negative at every horizon (frac_CI_neg = 1.00).

k	$\bar{\beta}_{\text{iso}}$	CV	$\bar{R}^2$	Null R^2	n_{cond}
1	−0.798	0.166	0.361	0.001	16
5	−0.710	0.146	0.281	0.001	16
10	−0.712	0.126	0.260	0.001	16
25	−0.703	0.143	0.226	0.002	16
50	−0.699	0.138	0.202	0.002	16
100	−0.701	0.137	0.182	0.001	16
200	−0.709	0.123	0.175	0.001	16

(diagonal-adjacency classes 1, 2, 4, 8; death fraction = 1.0 in GoL) die with near certainty at $t+1$, removing a local component-boundary node and reducing $C(t+1)$. *Multi-diagonal isolates* (classes with 2–4 diagonal neighbours) more often survive, but survive-connected: they merge with a larger component and again reduce the fine component count by absorbing an independent component. Both pathways lead to the same aggregate outcome—local fine-connectivity loss—which is why the local-window loss feature ($R^2 = 0.538$) dominates over simple death counts ($R^2 = 0.355$ for iso_count alone). The dominant process is the disappearance of fine-net connectivity in the local neighbourhood, aggregated over many isolate classes and many conditions.

TABLE II. Mechanism R^2 decomposition at horizon $k = 1$ (global pooled). CV: 5-fold cross-validation.

Feature	CV R^2	Direction
iso_count (prestate)	0.355	–
local_window_loss	0.538	–
all_fates (combined)	0.545	mixed

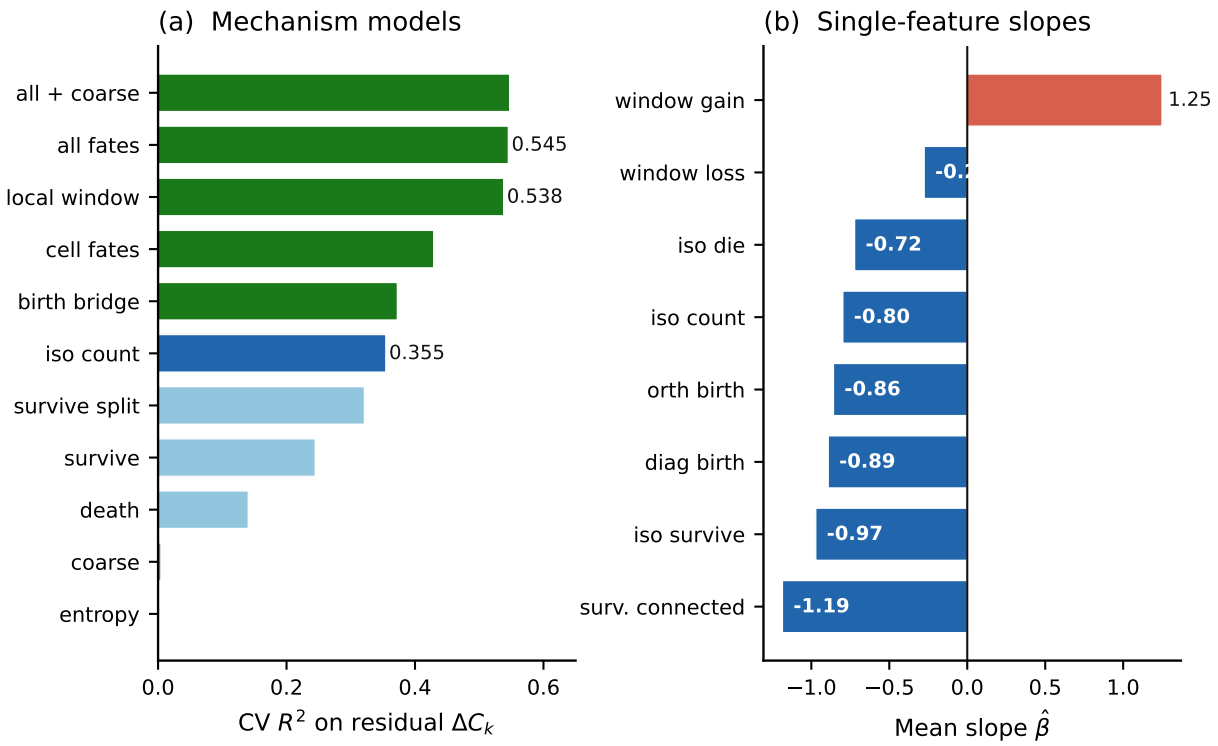

FIG. 5. **Result 4a: mechanism carrier decomposition.** (a) Cross-validated R^2 for each fate and mechanism feature. The local-window loss feature ($R^2 = 0.538$) outperforms iso_count alone ($R^2 = 0.355$), identifying local component-context loss as the mechanistic carrier. (b) Standardised slopes: loss-related features are negative, gain-related features are positive, revealing partial cancellation.

VII. RESULT 5 — TRANSFER, AMPLITUDE LAW, AND TWO-LAYER STRUCTURE

The mechanism analysis establishes what the carrier is. Two separate questions remain: does it *generalise* to held-out conditions, and does the *scale* of the response follow a systematic pattern? These are distinct questions because generalisation operates on the standardised direction of the slope vector, while amplitude concerns its raw magnitude. A mechanism can transfer perfectly in direction while having condition-specific amplitude; conversely, amplitude can be predictable even when directions differ. We address them in turn.

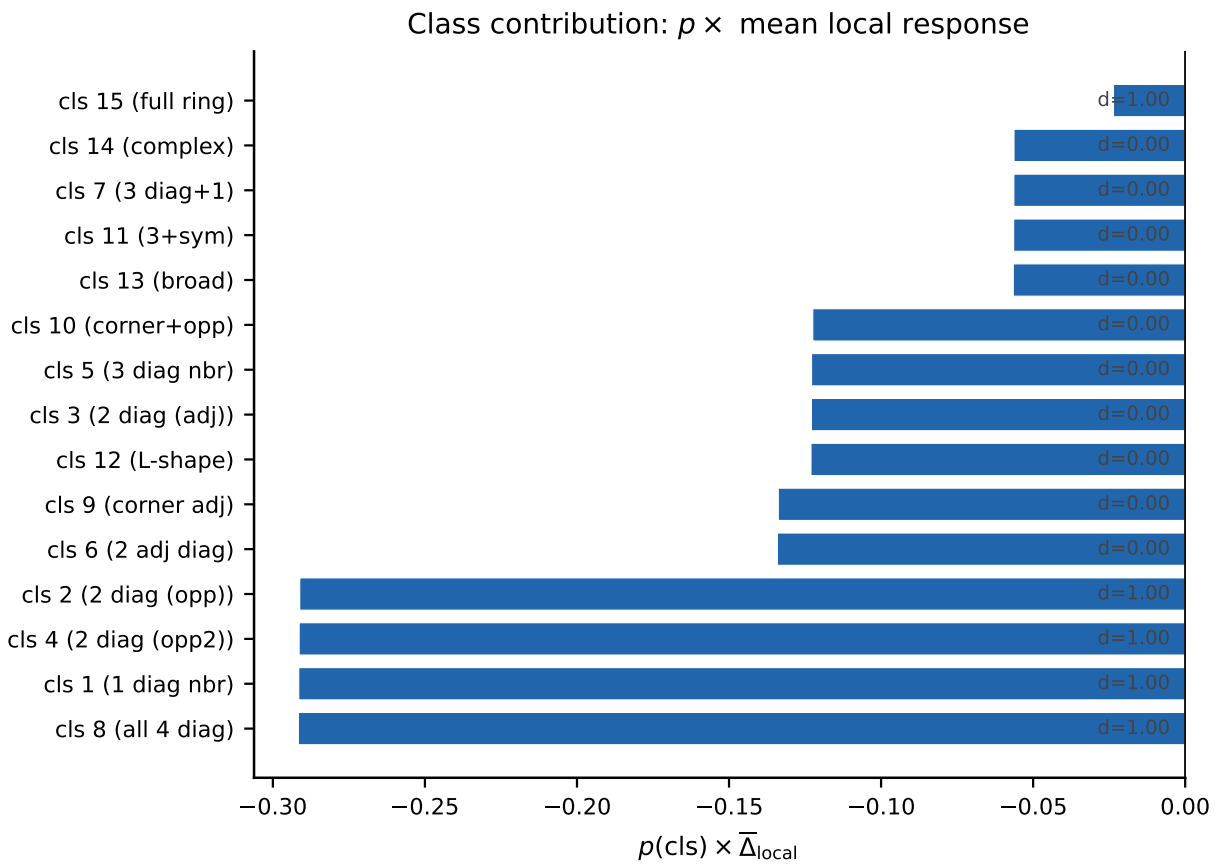

FIG. 6. **Result 4b: class-level contributions.** Horizontal bars show $p(\text{cls}) \times \overline{\Delta}_{\text{local}}$ for each diagonal-adjacency class. Annotations give the death fraction (d) from GoL transition-class statistics. All classes contribute negatively; single-diagonal-neighbour classes drive the bulk via their high prevalence.

A. Standardised Transfer

We z-score predictors and targets within each condition, then test whether the standardised mechanism coefficients transfer to held-out conditions under four leave-one-out axes (leave-density, leave-size, leave-rule, leave-condition). All four transfer axes pass (Fig. 7a). For the full fate representation, $R_z^2 = 0.544$ and frac R^2-positive $= 1.00$ on the leave-condition axis. iso_count alone also transfers ($R_z^2 = 0.355$). The standardised mechanism direction is condition-invariant: whichever leave-one-out split is applied, a model trained on the remaining conditions predicts the held-out condition well.

B. Amplitude Law

Condition amplitudes (raw residual slopes) vary substantially (amplitude CV = 0.369) but are predictable from (L, ρ). LOO R^2: full model = 0.977; size+ρ model = 0.970; size-only ≈ 0.802; rule-only = -0.138. Grid size L is the dominant predictor of amplitude; rule identity and density alone do not determine it.

The raw response thus decomposes as:

$$\hat{\beta}_{\text{iso}}(\text{condition}) \;\approx\; A(L, \rho)\, \hat{\beta}_{\text{iso}}, z\,, \tag{1}$$

where $\hat{\beta}_{\text{iso}}, z \;\approx\; -0.50$ is the transferable standardised slope and $A(L, \rho)$ is a condition amplitude predictable with LOO $R^2 \approx 0.977$. The natural explanation is that L sets the number of components and hence the denominator of the relative change $\Delta C_k/C(0)$, while ρ controls the density of the specific structural class; together they set the amplitude scale that multiplies the universal standardised direction.

VIII. TASK-DIRECTION COHERENCE

The results above establish a strong empirical picture. We now ask whether there is a formal geometric property of the fine-net task family that explains *why* a single iso_count statistic can serve all seven tested horizons simultaneously.

The diagnostic proceeds as follows. For each prediction task (a specific horizon k or a specific prediction target), we fit a residual regression model, then normalize the coefficient vector to unit length in whitened feature space. This gives a unit "task-direction vector" for each task. We then compute the pairwise absolute cosines $|\cos\theta_{ij}|$ across tasks within the same family, and the fraction of the total spread variance captured by the leading singular value (rank-1 energy).

A family with $|\cos| \approx 1$ and rank-1 energy ≈ 1 lies essentially along a single direction in description space: one feature set aligns all tasks simultaneously, which is precisely why a single statistic such as iso_count can serve the whole family. A family with $|\cos| \ll 1$ has geometrically dispersed tasks; no single feature can serve all of them.

For the fine-net horizon tasks, we find mean pairwise $|\cos| = 0.999$ and rank-1 energy = 0.999 (relative regret = 0.001). The fine-net family is essentially rank-1: a single description direction simultaneously aligns all k-step predictions. For the heterogeneous multi-target

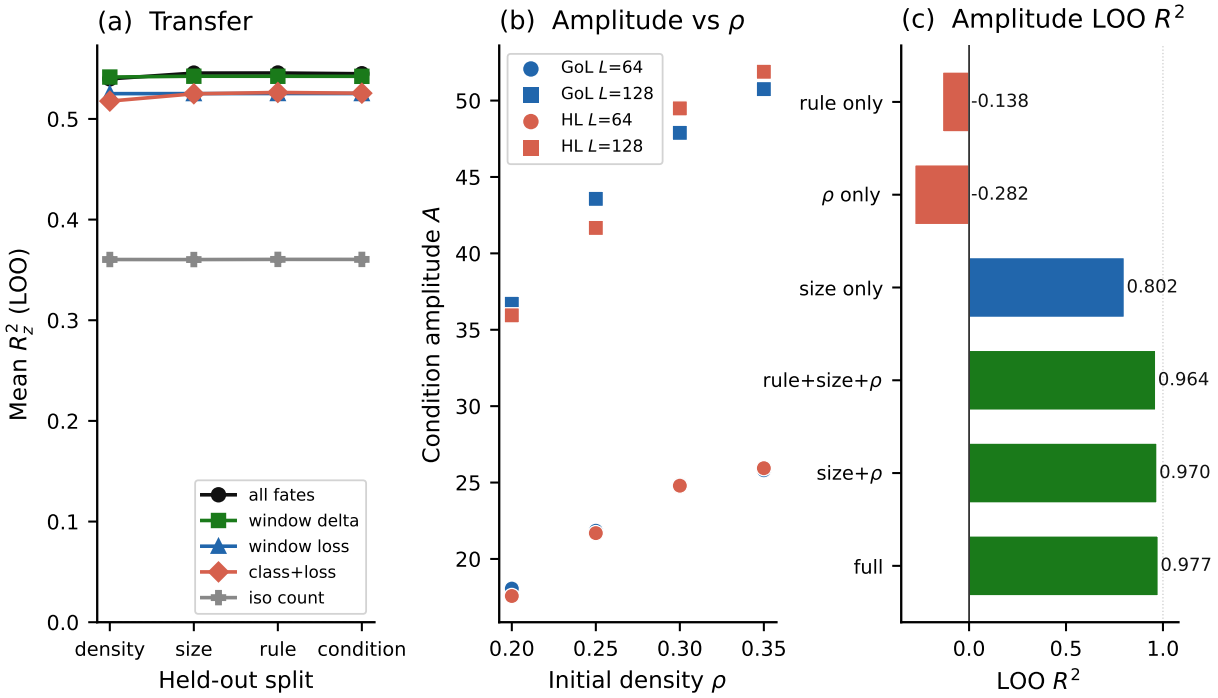

FIG. 7. **Result 5: transfer and amplitude law.** (a) Standardised transfer R_z^2 for five mechanism models across four leave-one-out axes; all lines stay positive. (b) Condition amplitude versus initial density ρ, coloured by grid size L and rule. Amplitude grows with L and is not determined by rule identity. (c) Amplitude LOO R^2 by model type; full model achieves $R^2 = 0.977$, size+ρ alone reaches 0.970.

battery, by contrast, mean pairwise $|\cos| = 0.427$, rank-1 energy $= 0.532$, and mean relative regret $= 0.596$. No single description direction serves all heterogeneous targets.

This contrast provides a compact geometric summary of the empirical results: iso_count works across all fine-net horizons because those tasks are nearly collinear in description space. The heterogeneous target family is not collinear, and indeed no single predictor adds $\Delta R^2 > 0.001$ for all targets simultaneously.

IX. DISCUSSION

What is earned. The five results form a coherent chain. *Selection (§III):* iso_count is a non-leaky target-specific predictor of fine-net component change, with $\Delta R^2 = 0.0012$ and incremental nulls near zero. *Prestate non-leakiness (§IV):* the full multi-horizon response is

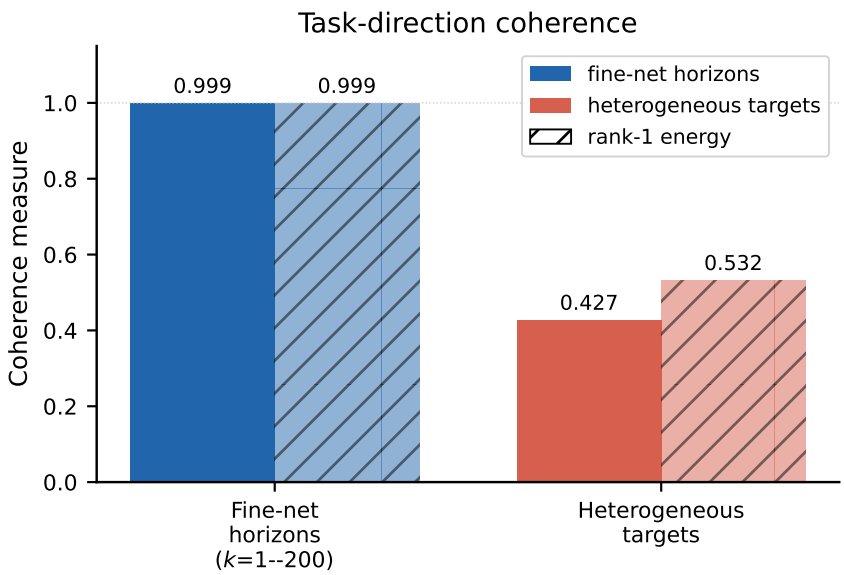

FIG. 8. **Task-direction coherence diagnostic.** Mean pairwise $|\cos\theta|$ (solid bars) and rank-1 energy (hatched bars) for fine-net horizon tasks (blue) and heterogeneous targets (orange). Fine-net horizons are near-perfectly aligned ($|\cos| = 0.999$, rank-1 energy $= 0.999$); heterogeneous targets are dispersed ($|\cos| = 0.427$, rank-1 energy $= 0.532$).

encoded in $t = 0$ iso_count alone; using fate information adds nothing substantial. *Temporal law (§V):* $\hat{\beta}_{\text{iso}}(k)$ is negative and stable across $k \in \{1, \ldots, 200\}$, both rules, both grid sizes, and all four density levels; no exceptions in 112 condition-horizon combinations. *Mechanism (§VI):* local component-context loss (not simple cell death) is the one-step carrier; the local-window feature accounts for 0.538 of the residual variance, compared with 0.355 for iso_count alone. *Transfer and amplitude (§VII):* the standardised mechanism transfers across all leave-one-out axes while the raw amplitude has a clean two-layer structure—transferable direction, predictable scale—with LOO $R^2 = 0.977$. *Coherence (§VIII):* fine-net horizon tasks are geometrically coherent ($|\cos| = 0.999$); heterogeneous targets are not ($|\cos| = 0.427$).

The two-layer structure. The decomposition $\hat{\beta}_{\text{iso}}(\text{cond}) \approx A(L, \rho)\, \hat{\beta}_{\text{iso}}, z$ cleanly separates the universal mechanism from the condition-specific scale. This matters for applications: if a practitioner has access to a new (L, ρ) condition, they can predict the raw slope amplitude with LOO $R^2 \approx 0.977$ from (L, ρ) alone, and they can apply the standardised slope direction directly from the training pool. The two layers are independently portable.

What is not earned. All results are scoped to {B3/S23, B36/S23} on periodic tori with

$L \in \{64, 128\}$ and $\rho \in \{0.20, 0.25, 0.30, 0.35\}$. Extension to other Life-like rules, larger grids, continuous-state CAs, or non-random initial conditions is an open empirical question. Closed-form derivation of $\hat{\beta}_{\text{iso}}$ from microscopic transition rules is not provided. Raw amplitude transfer without per-condition calibration is not supported: amplitude varies with (L, ρ) by a factor of ≈ 0.369 (CV).

Embedded isolates as objects. Embedded isolated cells are not just a convenient statistic: they form a well-defined structural class at the topological boundary of connectivity. Their prevalence determines how much the fine-component count is "at risk" at the next step. The stability of $\hat{\beta}_{\text{iso}}(k)$ across horizons suggests that the risk resolves primarily at $t = 1$ and that subsequent dynamics do not erase the initial predictive information—consistent with the prestate non-leakiness result. In this sense, iso_count is a *topological risk indicator* for fine-net change, rather than a simple density proxy.

X. CONCLUSION

The count of embedded isolated cells is the sufficient non-leaky prestate predictor of future fine-component change in Conway's Game of Life and HighLife. The response coefficient $\hat{\beta}_{\text{iso}}(k)$ is negative and stable across all 7 tested horizons and all 16 conditions (2 rules $\times$ 2 sizes $\times$ 4 density levels), constituting an empirical temporal law. Both iso-shuffle and target-shuffle incremental nulls are near zero ($\Delta R^2 < 0.001$), ruling out density artefacts. The one-step mechanism is local component-context loss, not simple cell death, and this standardised mechanism transfers across all leave-one-out axes. The raw response has a clean two-layer structure: a transferable standardised direction plus a predictable amplitude $A(L, \rho)$ (LOO $R^2 = 0.977$). Fine-net horizon tasks occupy a rank-1 coherent subspace of description space (mean pairwise $|\cos| = 0.999$), while heterogeneous prediction targets do not ($|\cos| = 0.427$), a compact geometric diagnostic of target-indexed description privilege.

ACKNOWLEDGMENTS

The author thanks the open-source communities behind NumPy, SciPy, scikit-learn, pandas, Matplotlib, and tqdm.

AUTHOR DECLARATIONS

Conflict of Interest

The author has no conflicts of interest to disclose.

Author Contributions

K.B.: conceptualization, methodology, software, investigation, formal analysis, writing.

DATA AVAILABILITY

All code, simulation outputs, and figure sources are available at `https://github.com/kunalb541/CA`.

Appendix A: Residualisation Protocol

For each prediction world at time t, the density nuisance covariate is $N_{\text{live}}(0)$, the total live-cell count of the initial state (proportional to ρ up to Bernoulli sampling variance). All regression models—both the selection-principle battery and the horizon response fits—include $N_{\text{live}}(0)$ as a linear covariate before adding iso_count or any fate feature. ΔR^2 is defined as $R^2(\text{density} + \text{predictor}) - R^2(\text{density only})$. The residual slope $\hat{\beta}_{\text{iso}}$ is the OLS coefficient on iso_count in this density-residualised fit. Within each condition the density range is narrow (ρ varies by at most sampling noise around its target value), so the density covariate absorbs only small amounts of within-condition variance; its primary role is to enable clean pooling across conditions in the global summary.

Appendix B: Null Definitions

Two nulls are computed: (i) *iso-shuffle null*: iso_count labels are permuted uniformly at random across worlds within each condition; ΔR^2 and slope are recomputed on the permuted predictor. (ii) *target-shuffle null*: the target variable ΔC_k is permuted across worlds; ΔR^2 is recomputed with the original iso_count. Both nulls converge near zero ($\Delta R^2 < 0.001$),

confirming that the signal depends on the joint alignment of iso_count values with ΔC_k values and is not recoverable by a random permutation of either.

Appendix C: Transition-Class Coding

Diagonal-adjacency class is an integer in $[1, 15]$ indexing the bit pattern of the four diagonal neighbours of an embedded isolate (NW, NE, SW, SE), with class c equal to the binary value $NW \cdot 1 + NE \cdot 2 + SW \cdot 4 + SE \cdot 8$. Class 15 (all four diagonal neighbours alive) is rare in this battery ($p \approx 0.009$, $n = 80{,}015$ instances) but contributes a negative local response ($\bar{\Delta} \approx -2.52$, death fraction = 1.0) comparable to other multi-diagonal classes; it is not excluded from the analysis. Classes 1, 2, 4, 8 are the *single-diagonal* classes; classes 3, 5, 6, 9, 10, 12 are the *two-diagonal* classes; classes 7, 11, 13, 14 are the *three-diagonal* classes; class 15 is the *four-diagonal* class.

Appendix D: Test and Regression Battery

The repository includes 90 automated tests in `test_regression.py` covering:

- correctness of the BFS component counter and isolate classifier in `ca.py`;
- selection-principle verdicts (GoL PASS, Global PASS);
- horizon response properties (all slopes negative, all CI negative, R^2 above floor, null near zero);
- fate mechanism ordering (local-window > iso_count > coarse);
- standardised transfer (frac R^2-positive = 1.0);
- amplitude-law LOO R^2 thresholds;
- LGDS / task-direction coherence values;
- prestate non-leakiness across all horizons;
- artifact generator checks (no NaN macros, no digit-containing macro names, all figure PDFs present).

All 90 tests pass against the committed `outputs/` CSV files.

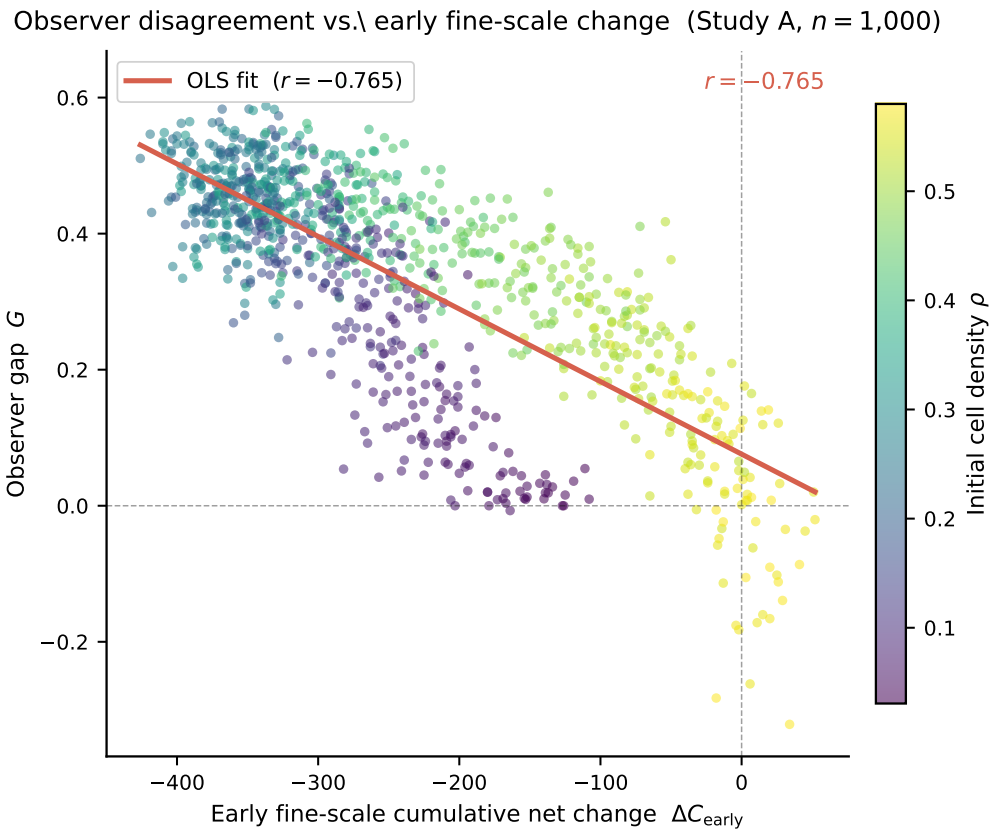

FIG. 9. **E1 — Observer disagreement scatter (Study A, $n = 1{,}000$).** Each point is one simulated GoL world. x-axis: early fine-scale cumulative net component change ΔC_{early}; y-axis: observer gap G (fine minus coarse observer agreement); colour: initial cell density ρ. The regression line gives $r = -0.765$ ($p \ll 10^{-100}$); partial r controlling for density $= -0.848$. Worlds where fine-scale structure degrades early tend to produce larger observer disagreement. *Background diagnostic only; not load-bearing for the response-law claims in the main text.*

Appendix E: Background Observer-Scale Diagnostics

The main text establishes the embedded-isolate response law using the 16-condition GoL/HighLife battery described in Section II. For lineage and context, this appendix records earlier observer-scale diagnostics from the same research program. These diagnostics show *why* a target-indexed description analysis was a natural next step: fine and coarse observers disagree systematically, and different targets are optimally predicted at different descriptive scales. **These figures are background diagnostics only; they are not load-bearing evidence for the response-law estimates in the main text.**

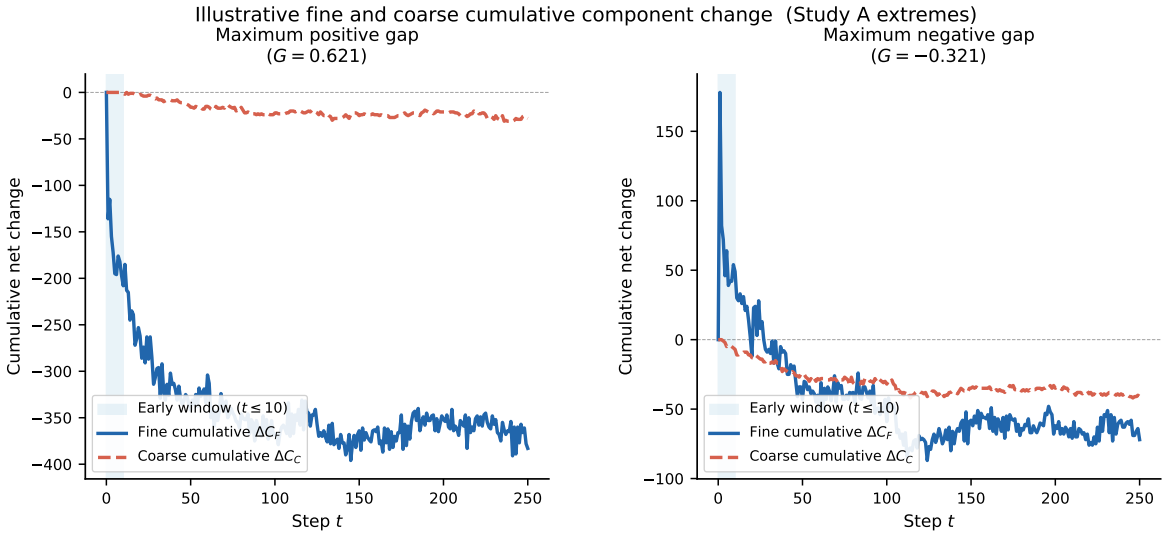

FIG. 10. **E2 — Illustrative fine and coarse observer trajectories (Study A extremes).** Left: the world with the maximum positive observer gap ($G = +0.621$). Right: the world with the maximum negative gap ($G = -0.321$). Blue solid: fine-scale cumulative net component change $\Delta C_F(t)$. Orange dashed: coarse-scale cumulative net change $\Delta C_C(t)$. Shaded region: early window ($t \leq 10$) whose summary statistic ΔC_{early} is used in Figure 9. These are selected extreme cases, not representative samples. *Background diagnostic only; not load-bearing for the response-law claims in the main text.*

REFERENCES

[1] J. C. Flack, "Coarse-graining as a downward causation mechanism," Philosophical Transactions of the Royal Society A **375**, 20160338 (2017).

[2] C. R. Shalizi and C. Moore, "What is a macrostate? Subjective observations and objective dynamics," arXiv preprint cond-mat/0303625 (2003).

[3] E. P. Hoel, L. Albantakis, and G. Tononi, "Quantifying causal emergence shows that macro can beat micro," Proceedings of the National Academy of Sciences **110**, 19790–19795 (2013).

[4] M. Gardner, "Mathematical games: The fantastic combinations of John Conway's new solitaire game "life"," Scientific American **223**, 120–123 (1970).

[5] C. G. Langton, "Computation at the edge of chaos: Phase transitions and emergent computation," Physica D: Nonlinear Phenomena **42**, 12–37 (1990).

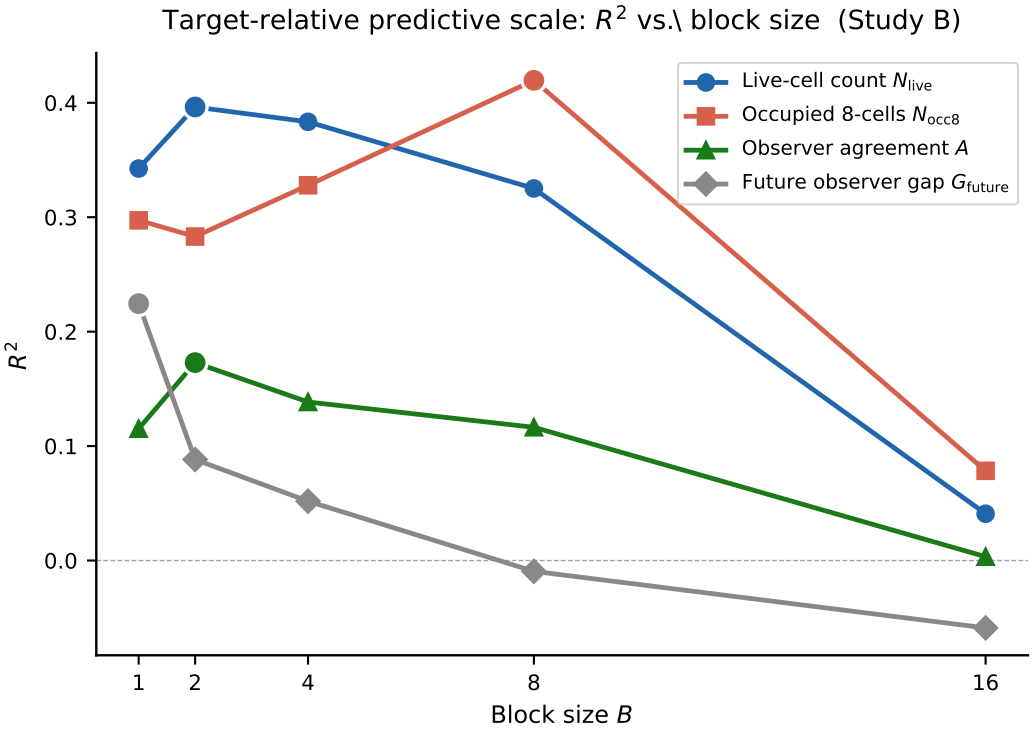

FIG. 11. **E3 — Target-relative predictive scale (Study B).** R^2 of a block-aggregated feature predictor as a function of block size B, separately for four prediction targets. N_{live} peaks at $B = 2$; N_{occ8} peaks at $B = 8$; future observer gap G_{future} peaks at $B = 1$ and falls steeply; observer agreement A is broadly flat then drops. Different targets favour different descriptive scales, motivating the target-specific framing of the response-law analysis in the main text. *Background diagnostic only; not load-bearing for the response-law claims in the main text.*

[6]S. Wolfram, "Universality and complexity in cellular automata," Physica D: Nonlinear Phenomena **10**, 1–35 (1984).

[7]J. P. Crutchfield and K. Young, "Inferring statistical complexity," Physical Review Letters **63**, 105–108 (1989).

[8]A. Rupe and J. P. Crutchfield, "Local causal states and discrete coherent structures," Chaos **28**, 075312 (2018).

[9]N. Israeli and N. Goldenfeld, "Computational irreducibility and the predictability of complex physical systems," Physical Review Letters **92**, 074105 (2004).

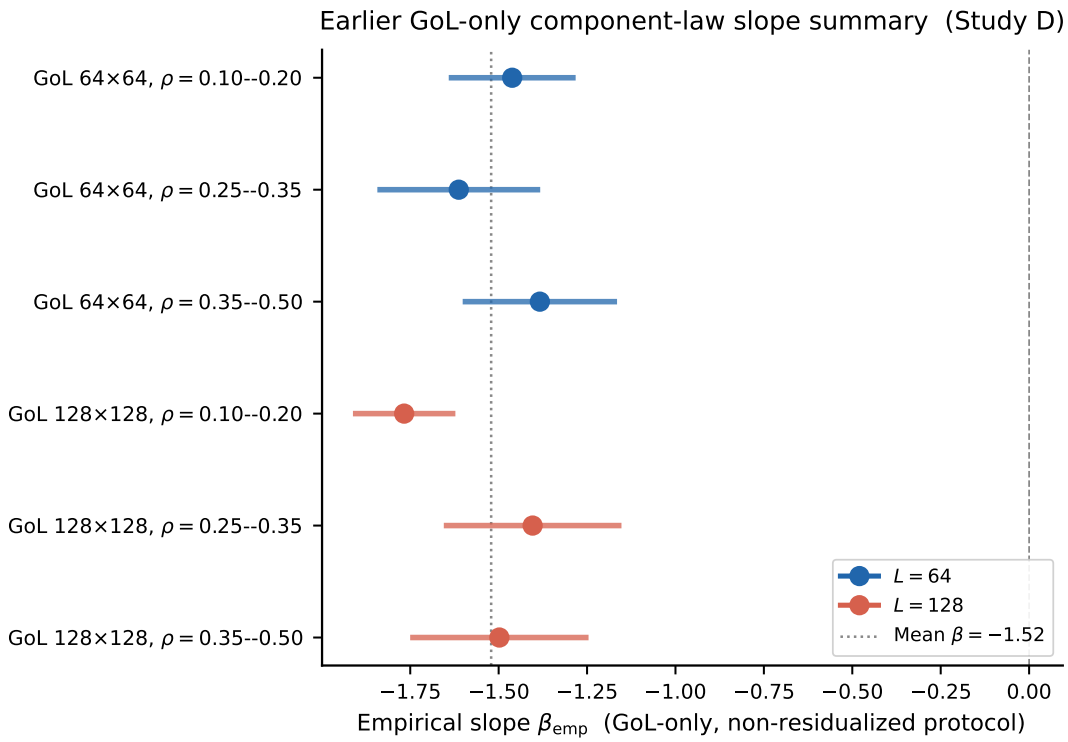

FIG. 12. **E4 — Earlier GoL-only component-law slope summary (Study D).** Empirical slopes β_{emp} with 95% CI for six GoL conditions (two grid sizes, three density bands) under an earlier, non-residualized regression protocol. Mean $\beta \approx -1.52$. **This coefficient is not directly comparable to the residualized horizon slopes $\beta_{\text{iso}} \approx -0.70$ in the main text:** the two studies use different targets, different residualization, and different simulation batteries. This figure is included as historical lineage only. *Background diagnostic only; not load-bearing for the response-law claims in the main text.*